

Game of Life

CPU vs CUDA

One algorithm. 100 generations. Two execution environments.

Upload CUDA JSON

Load CPU JSON

Clear CUDA

Download comparison CSV

CUDA results loaded. 5 matched grid sizes.

CPU runs on your local Mac; CUDA runs on a Colab NVIDIA GPU. Speedup compares these complete configurations, including different hardware and compilers. It does not isolate the effect of parallelization.

Iterations

100

CPU grids

5

CUDA grids

5

Peak speedup

93.353x

Experiment

Local CPU

Apple M1

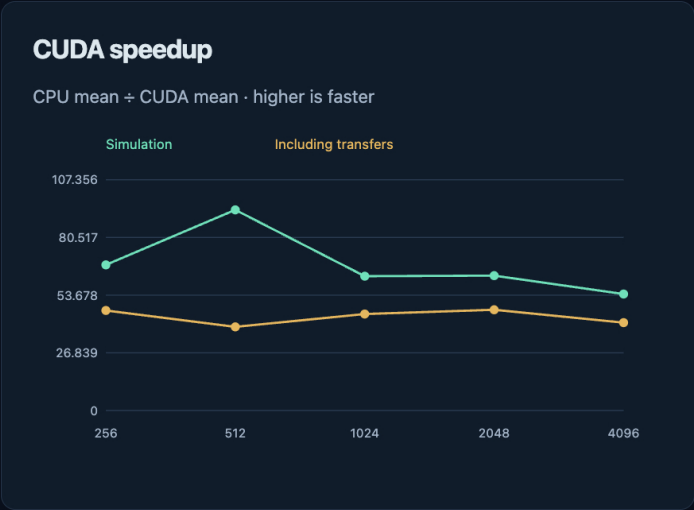
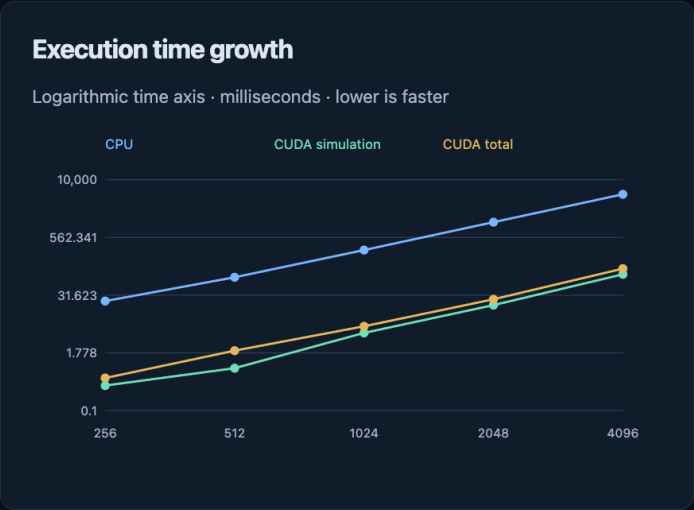
3 runs · 2026-10-07T14:30:31.017574+00:00

Colab CUDA

Tesla T4

3 runs · 2026-10-07T14:33:15.278637+00:00

Seed 42 · dead exterior · contiguous grids · three runs per size · 100 iterations per run. Simulation times exclude initialization, allocation, validation and printing. CUDA total includes transfers and synchronization, excluding allocation and warm-up.



Measurements

Mean ± sample standard deviation. Speedup is shown only for compatible results with matching final-grid SHA-256 fingerprints.

Grid / cells	Iterations	CPU ms	CUDA ms	CUDA total ms	Speedup	Total speedup	Result check
256 x 256 / 65,536	100	23.511 ± 5.861	0.347 ± 0.011	0.505	67.765x	46.531x	SHA-256 match

512 × 512 / 262,144	100	77.065 ± 0.416	0.826 ± 0.004	1.98	93.353x	38.916x	SHA-256 match
1024 × 1024 / 1,048,576	100	299.852 ± 1.068	4.798 ± 0.5	6.681	62.494x	44.884x	SHA-256 match
2048 × 2048 / 4,194,304	100	1,205.305 ± 1.164	19.206 ± 0.003	25.715	62.758x	46.871x	SHA-256 match
4096 × 4096 / 16,777,216	100	4,846.51 ± 43.354	89.437 ± 12.01	118.552	54.189x	40.881x	SHA-256 match

Measured insights

- Highest observed simulation speedup: 93.353x at 512 × 512.
- First tested grid reaching 2x simulation speedup: 256 × 256. Here “significant” means a descriptive 2x threshold, not statistical significance.
- From 256 to 4096, cells increase 256x; CPU time grows 206.135x and CUDA simulation time grows 257.777x.
- Speedup is not monotonic across matched tested sizes.
- At the smallest matched grid, CUDA total is 0.505 ms versus 0.347 ms simulation time. The difference includes transfers and host/synchronization overhead; these timings do not isolate its individual causes.

ALGORITHM DEMONSTRATION

Watch the generations evolve

Live cells survive with 2–3 neighbours; dead cells are born with 3. All other cells die or stay dead. Each update reads the previous generation.

This 48 × 48 browser simulation is JavaScript, not a benchmark or CUDA execution.

StartPauseStepResetRandomize

Generation 0



How the implementations work

CPU / sequential loops

One host thread traverses rows and columns in contiguous system memory. Two buffers preserve simultaneous updates; pointer swaps avoid grid copies. The compiler may vectorize work.

CUDA / parallel cells

Each valid CUDA thread updates one cell. 16 × 16 blocks access device memory. Both buffers stay on the GPU for all 100 launches. Blocks and warps are scheduled over the available GPU hardware.